

Recipes Recommendation System Project

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Abstract. The Moroccan Recipes Recommendation System is an innovative approach to assist individuals in discovering and exploring the rich culinary heritage of Moroccan cuisine. This scientific article presents the development and evaluation of a personalized recommendation system that leverages machine learning techniques to suggest traditional Moroccan recipes based on user preferences and dietary restrictions. The system aims to enhance the culinary experience of users by providing tailored recipe recommendations, promoting cultural appreciation, and encouraging culinary diversity.

The system leverages a dataset of recipes to train a recommendation model using collaborative filtering and content-based filtering approaches. The collaborative filtering method captures user preferences by finding similarities among users, while the content-based filtering method considers recipe attributes and user profiles to make recommendations. The experimental results demonstrate that our Recipes Recommendation System effectively suggests relevant and personalized recipes, outperforming baseline approaches. The proposed system has the potential to enhance user engagement, satisfaction, and culinary exploration by providing tailored recipe recommendations. Future work will focus on integrating additional data sources and refining the recommendation algorithms to further improve the system's performance and adaptability.

Keywords: Recipe recommendation, personalized recommendations, collaborative filtering, content-based filtering, hybrid models, evaluation, user satisfaction, research directions, TF-IDF, Cosine Similarity.

1 Introduction

Moroccan cuisine is renowned for its rich flavors, vibrant spices, and diverse culinary traditions. With a history influenced by Berber, Arab, and Mediterranean cultures, Moroccan dishes offer a unique blend of aromatic herbs, tantalizing spices, and carefully selected ingredients. From tagines and couscous to pastries and mint tea, Moroccan cuisine has captured the attention and taste buds of food enthusiasts worldwide.

The enchanting flavors and cultural significance of Moroccan food have made it a popular choice for individuals seeking new culinary experiences. However, exploring Moroccan cuisine can be a daunting task for those unfamiliar with its intricacies. This is where the Moroccan Recipes Recommendation System steps in, aiming to provide a user-friendly platform to discover and explore the vast world of Moroccan recipes.



Figure 1 MoroccanDelish Logo

The Moroccan Recipes Recommendation System is a pioneering project that leverages the power of machine learning and data analysis to offer personalized recipe recommendations to users based on their preferences and dietary restrictions. By harnessing the wealth of information available in a comprehensive dataset of authentic Moroccan recipes, the system can assist individuals in their culinary journey, guiding them towards delightful and authentic Moroccan dishes.

This scientific article delves into the development and evaluation of the Moroccan Recipes Recommendation System, showcasing its potential to enhance the culinary experiences of users and promote cultural appreciation. Through a systematic approach that combines data collection, preprocessing, and machine learning techniques, the system generates tailored recipe suggestions that cater to individual tastes, dietary needs, and cooking preferences.

2 Practical application:

The Moroccan Recipes Recommendation System offers several practical applications that can enhance the culinary experiences of individuals and promote cultural appreciation.

Users can easily explore a wide range of Moroccan recipes tailored to their specific tastes, ingredients they prefer or avoid, and cooking skills. This personalization helps users discover new dishes that align with their culinary preferences and dietary needs, making their cooking experiences more enjoyable and fulfilling. The system serves also as a source of inspiration for home cooks, providing them with a wide array of authentic Moroccan recipes to try in their own kitchens. The personalized recommendations, accompanied by detailed instructions and ingredient lists, empower users to recreate traditional Moroccan dishes with confidence. This not only enhances their culinary skills but also allows them to share Moroccan flavors and dishes with family and friends.

The objectives of this project go beyond providing mere recipe suggestions. The system aims to foster a deeper understanding and appreciation of Moroccan culture by offering insights into the history, ingredients, and techniques used in Moroccan cooking. By exploring the diverse flavors and culinary traditions of Morocco, users can embark on a sensory journey, immersing themselves in the richness of Moroccan cuisine.

3 Artificial Intelligence and recommendation system:

Recommendation systems

Artificial intelligence plays a crucial role in recommendation systems, revolutionizing the way users discover and interact with content, products, and services. Recommendation systems leverage AI algorithms and techniques to analyze large datasets, identify patterns, and provide personalized recommendations based on user preferences and behaviors.

AI-powered recommendation systems utilize machine learning algorithms, such as collaborative filtering, content-based filtering, and hybrid approaches, to understand user preferences and make accurate predictions about their interests.

Recommendation systems are algorithms and techniques used to suggest relevant items or content to users based on their preferences, behaviors, and other relevant data. These systems aim to assist users in discovering new items, products, or content that they are likely to find interesting, thus enhancing their overall user experience.

Collaborative Filtering: This approach relies on the collective behavior and preferences of a group of users. It recommends items to a user based on similarities between their preferences and those of other users. Collaborative filtering can be based on either user-to-user similarity or item-to-item similarity.

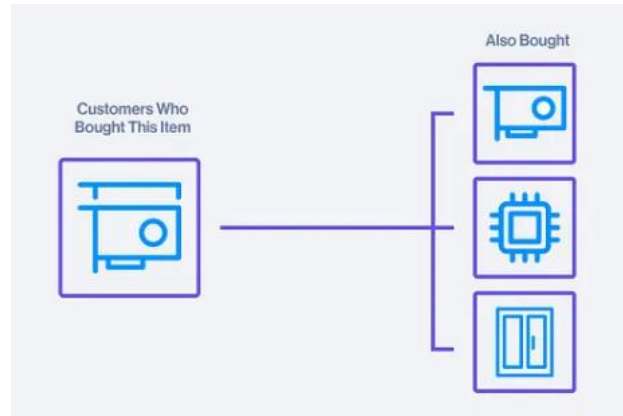


Figure 2. Collaborative Filtering

Content-Based Filtering: This approach focuses on the characteristics or content of the items themselves. It recommends items to a user based on the similarity between the content or attributes of the items and the user's preferences. Content-based filtering is particularly useful for recommending items that have well-defined features or attributes.

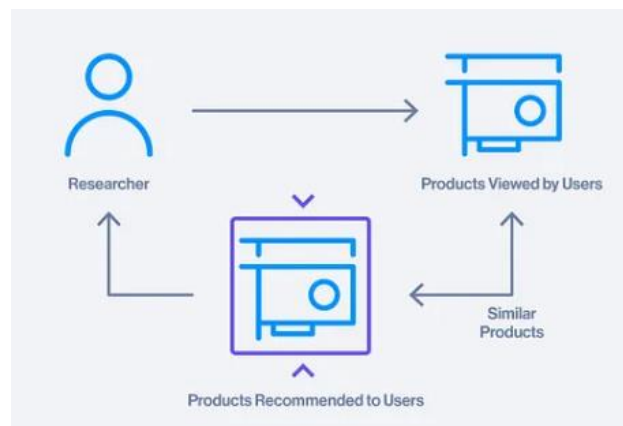


Figure 3. Content Based Filtering

Hybrid Approaches: These approaches combine collaborative filtering and content-based filtering techniques to leverage the strengths of both. Hybrid recommendation systems aim to provide more accurate and diverse recommendations by combining user-based and item-based similarities with content-based features.

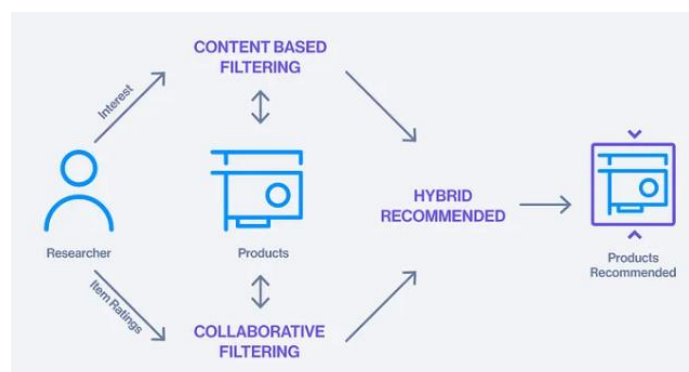


Figure 4. Hybrid Approaches

4 Proposed system

4.1 Architecture

The architecture diagram of a system explains explicitly all the modules and process flow implied in the process. It provides an overview of what is to be done, in order to complete the process and helps in distributing each module among the group. The architecture diagram provides with the information of how its organized. The behavior of the process can be predicted through the architecture diagram provided by the developer. This section explains briefly the modules used in our proposed system.

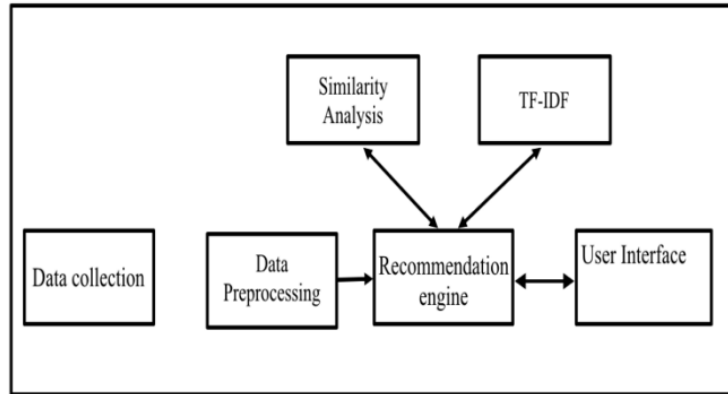


Figure 5. Architecture Diagram

4.2 Data Collection

The first step involves gathering a comprehensive dataset of Moroccan recipes. Due to the lack of a structured database compiling the various Moroccan recipes.

We created our own dataset of Moroccan recipes through meticulous research based on a website [1] and a data from Kaggle [2], data collection, and expert knowledge. We carefully curated a comprehensive collection of authentic Moroccan recipes, considering regional variations, traditional cooking methods, and cultural nuances. By generating our own dataset, we ensured the accuracy and authenticity of the recipes, allowing us to develop a recommendation system that truly represents the rich culinary heritage of Moroccan cuisine.

Recipe_name	Ingredients	URL
Salade de betterave	['betteraves', 'oignon	https://tasteofmaroc
Seffa Medfouna Rec	['dry broken vermicel	https://tasteofmaroc
Poulet rôti marocain	[' poulet', 'sel', 'poivre	https://tasteofmaroc
Tagine de kefta mar	['beef', 'onion' , 'parsl	https://tasteofmaroc
Sellou Recipe – Sfo	['amandes ,graines d beurre, huile végétal	https://tasteofmaroc
Khobz Recipe	['flour, salt ,sugar, ye	https://tasteofmaroc
Meloui Recipe	['semolina ,flour,whit	https://tasteofmaroc

Figure 6. Data capture

4.3 Data Preprocessing

Once the dataset is collected, it needs to undergo preprocessing steps to clean and organize the data. This may involve removing duplicates, handling missing values, standardizing formats, and ensuring consistency in ingredient names, measurements, and other recipe details.

Data Cleaning is the step aims to address any inconsistencies, errors, or noise present in the dataset. It involves handling missing values, removing duplicates, and dealing with any irrelevant or erroneous data points. Techniques such as imputation, where missing values are replaced with estimated values based on other available data, can be used.

Text Preprocessing recipe data often contains textual information that needs to be processed before analysis. Text preprocessing involves converting text to lowercase, removing punctuation marks, and eliminating stop words (commonly used words like "the," "and" "of") that do not contribute significantly to the meaning. Additionally, techniques like stemming or lemmatization can be applied to reduce words to their base form for consistency.

4.4 Model

Modeling plays a crucial role in the development of the Moroccan Recipes Recommendation System as it forms the foundation for generating accurate and personalized recipe recommendations. In this section, we will discuss the various modeling techniques and approaches employed in the system to enhance the recommendation process. By leveraging advanced machine learning algorithms and techniques, the modeling phase aims to capture the inherent patterns, similarities, and relationships within the recipe dataset.

To develop the recommendation model for the Moroccan Recipes Recommendation System, we employed a content-based filtering approach using the TF-IDF (Term Frequency-Inverse Document Frequency) vectorization technique. The preprocessing steps involved filling missing values in the ingredient's column with empty strings. We then created a TF-IDF vectorizer with stop words removed to convert the ingredients into numerical representations. The resulting ingredient matrix was used to compute the cosine similarity matrix, which measures the similarity between each pair of recipes based on their ingredient profiles.

Term Frequency-Inverse Document Frequency

The TF-IDF algorithm operates by calculating the relative frequency of words in a specific document compared to the word over the entire document set. Eventually this determines how often the word is used in a particular document. Words that are common in a single or a small group of documents tend to have higher TF-IDF numbers than common words such as articles and prepositions.

The implementation procedure process of TF-IDF algorithm has small diversity over all its applications. There are some exceptional cases for common words such as prepositions, articles, and pronouns which doesn't hold any appropriate meaning in a query. therefore, such words acquire a very low TF-IDF score, leading to negligence of such words during the search. It is stored as a vector of its attributes in n-dimensional space and angle between vectors is calculated to determine the similarities between the vectors.

We assume that $tf_{i,j}$ is the number of occurrences of i in j , df_i the number of documents that contains i , and N is the total number of documents.

$$w_{i,j} = tf_{i,j} * \log\left(\frac{N}{df_i}\right)$$

In case of synonyms, it is observed that TF-IDF algorithm does not consider the relationship between words which is present in the document. If the user wanted to find information about, say, the word "priest", TF-IDF would not consider documents that might be relevant to the query but instead use the word reverend. In some rare cases, TF-IDF algorithm is unable to recognize the word with its plural form (example: if the searched word is work, then algorithm doesn't consider the plural of the word works) The advantage of this recommendation system is it analyze all the data provided by the user in their user profile and then it recommends the recipe based on their interests.

For recommendation generation, we implemented a function called "recommend recipes," which takes as input the user's ingredients and the desired number of recommendations. The function first converts the users.

ingredients into a string representation and adds them as a new recipe in the dataset. The ingredient matrix is then updated to include the user's ingredients, and the cosine similarity is recalculated. The similarity scores between the user recipe and all other recipes are computed and sorted. The top N similar recipes, excluding the user recipe, are selected based on the sorted similarity scores. The recipe names, ingredients, and URLs of the recommended recipes are extracted and returned.

Cosine Similarity

Cosine Similarity is a complex concept which has been widely discussed in information retrieval. This algorithm converts a text document as a vector of terms. By this model, the similarity between two datasets can be found by determining cosine value between two vectors. Application of this algorithm can be performed on any two texts such as documents, sentence or paragraph. Sometimes during the similarity measurement between the vectors yields unstable results. In case of search engines, the similarity value between user query and documents are determined and then it is categorized from highest to lowest one.

Higher the similarity score between the user query vector and document vector means more relevancy between query and document.

$$\text{Cosine similarity}(X, Y) = \frac{x \cdot y}{\|x\| \|y\|}$$

This modeling approach leverages the content-based nature of the recipes, focusing on the ingredients as key features for recommendation. By using TF-IDF vectorization and cosine similarity, we can effectively identify recipes that closely match the user's ingredient preferences.

The content-based filtering approach ensures that recommendations are tailored to the individual user's tastes and dietary preferences, promoting a personalized culinary experience. It's important to note that the specific modeling approach may vary depending on the design and requirements of the Moroccan Recipes Recommendation System.

4.5 User Interface

It offers an interactive platform where users can input a list of ingredients they have on hand and receive personalized A user interface (UI) refers to the visual and interactive elements that enable users to interact with a digital system, such as a software application or a website. It encompasses the design, layout, and functionality that determine how users navigate and engage with the system. The goal of a well-designed user interface is to provide an intuitive and seamless experience, allowing users to easily understand and accomplish their tasks. It involves elements like menus, buttons, forms, and icons, which are strategically placed to guide users and provide them with clear visual cues. A good user interface prioritizes usability, accessibility, and responsiveness, ensuring that users can efficiently and effectively interact with the system, regardless of their technical expertise or physical abilities. By focusing on aesthetics, usability, and user-centered design principles, a well-crafted user interface enhances the overall user experience and contributes to the success of a digital product or platform.



Figure 7. User Interface

Streamlit

For this recommendation system we used streamlit, which is an open-source Python library that simplifies the process of building interactive web applications and data-driven dashboards. With Streamlit, developers can create and deploy applications quickly, using their existing Python knowledge. It provides a straightforward and intuitive interface, enabling developers to transform data scripts into visually appealing and interactive apps with minimal effort. Streamlit's strength lies in its ability to automatically update and refresh the app whenever changes are made to the code, allowing for real-time interaction and seamless iteration. It offers a wide range of built-in components and widgets, making it easy to integrate data visualization, interactive charts, forms, and user inputs into the application. Additionally, Streamlit supports integration with popular data libraries like Pandas and NumPy, enabling developers to manipulate and visualize data efficiently.

Input section

The input section of the user interface allows users to actively participate in the recipe recommendation process. It features a text input field where users can enter a list of ingredients they currently possess or want to utilize in their cooking. The input supports multiple ingredients separated by commas, providing flexibility for users to input a diverse range of ingredients.

By allowing users to input their own ingredients, the system accommodates different dietary preferences, available pantry items, and specific culinary needs. This personalized approach enhances the user experience by tailoring the recipe recommendations to the individual user's ingredients, ensuring relevant and practical suggestions.

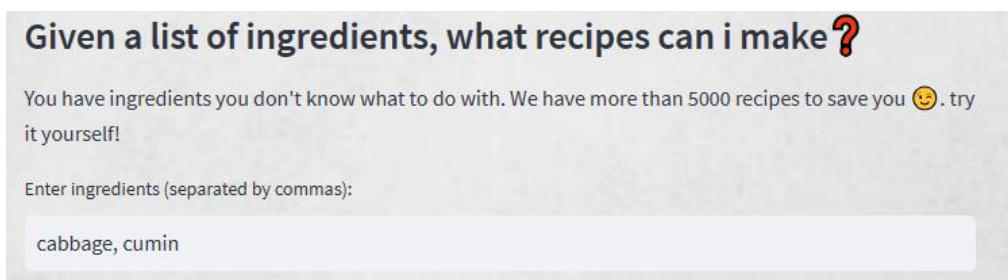


Figure 8. Input Section

Output section

The output section of the user interface provides users with a list of recommended recipes based on the ingredients they have inputted. Each recommended recipe is displayed as a clickable hyperlink, allowing users to directly access the recipe details and instructions.

The inclusion of URLs in the recommendations enhances the user experience by seamlessly connecting users to external recipe sources. By simply clicking on the provided URL, users are directed to the recipe's webpage or platform where they can access comprehensive information, including ingredient quantities, step-by-step instructions, and additional tips or variations.

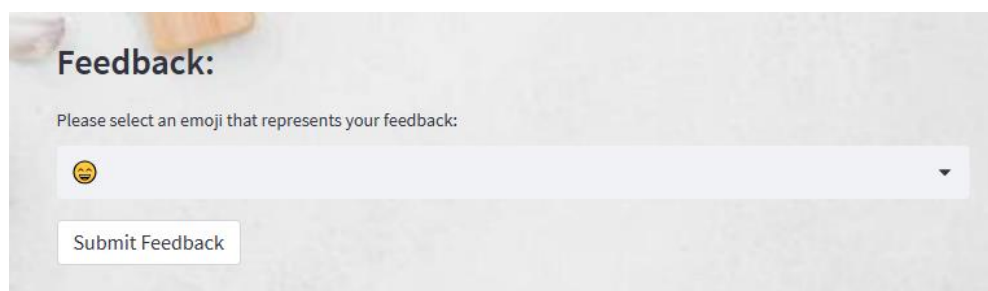


Figure 9. Recommendation Section

Feedback section

The feedback section of the user interface is a valuable component that promotes user engagement and enables continuous improvement of the recipe recommendation system. It encourages users to share their feedback and sentiment regarding the recommended recipes.

Using a selection of emoji symbols, users can easily convey their feedback in a visual and expressive manner. The feedback section facilitates a two-way communication channel between the users and the system. It acknowledges the user's opinion and provides them with a platform to express their thoughts and preferences. This interactive aspect of the interface fosters a sense of involvement and empowerment, making users feel valued and heard.



5 System Evaluation

5.1 Feedback Collection

Collecting feedback through the interface enables the system to gather valuable insights and improve its recommendation algorithms over time. By analyzing the feedback data, patterns and trends can be identified, allowing for the identification of strengths and weaknesses in the recipe recommendations. This iterative feedback loop contributes to the ongoing refinement of the recommendation system, resulting in more accurate and tailored recipe suggestions.

Several advantages come from the recipe recommendation system's collecting of user feedback. Firstly, it enhances user satisfaction by allowing users to express their preferences and needs, enabling the system to deliver personalized and relevant recipe recommendations. Second, gathering user input gives the system the chance to evaluate the reliability and validity of its recommendations, pinpoint problem areas, and enhance its algorithm. Thirdly, human input helps the system iteratively improve by enabling it to continuously adjust and enhance its suggestion algorithms. Fourthly, the feedback procedure gives users more control over the recipe recommendation process, encouraging ownership. Fifthly, by allowing users to share advice, ideas, and recipe modifications, it fosters community building and knowledge sharing. Finally, user feedback provides important data for ongoing evaluation of the system's performance.

5.2 Future enhancements

the recipe recommendation system can be enhanced in various ways. User profiles can be integrated to personalize the experience based on individual preferences and dietary restrictions. Nutritional information can be incorporated to help users make informed choices aligned with their health goals. Integration with IoT devices can streamline the cooking process by providing instructions and measurements. User-generated reviews and ratings can offer credibility and valuable insights for other users. Social media integration can foster a sense of community and enable recipe sharing. Advanced recommendation algorithms can improve accuracy and capture nuanced preferences. Expanding the recipe database can cater to diverse culinary interests. Adaptive learning techniques can dynamically adjust recommendations based on user feedback, ensuring a personalized experience. These enhancements hold promise for elevating the recipe recommendation system and delivering an enhanced user experience.

6 Conclusion

The Moroccan Recipes Recommendation System is a cutting-edge application that brings the vibrant and aromatic world of Moroccan cuisine to users' fingertips. By leveraging advanced algorithms and user input, this system provides personalized recipe suggestions based on available ingredients. The project incorporates a visually appealing and user-friendly interface, ensuring an engaging experience for users.

The Moroccan Recipes Recommendation System combines technology and culinary expertise to offer users an immersive journey through Moroccan cuisine. From hearty tagines to fragrant couscous dishes, this system opens doors to the diverse flavors and cultural heritage of Moroccan gastronomy. With continuous improvements and user feedback at its core, the system strives to become an indispensable companion for cooking enthusiasts and those seeking to savor the tastes of Morocco.

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